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Combining GPS Ionospheric Seismology and Raytracing Technique to Detect Acoustic Waves in Near-field Triggered by the 11 March 2011 Tohoku Earthquake

Pil Hun Choi and Jiyun Lee

Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, Daejeon,
Republic of Korea

phchoi@kaist.ac.kr

Introduction

The Acoustic-Gravity Waves (AGWs) triggered by vertical ground displacement such as earthquakes, tsunamis and underground nuclear tests reach to the ionosphere. Previous works have observed and analyzed signatures of AGWs-generated perturbations in the ionosphere using GNSS remote sensing technique to understand the mechanism of the dynamic coupling. In this work, we process GPS measurements obtained from GNSS stations in Japan and analyzed Coseismic Ionospheric Disturbances (CIDs) that were induced by 11 March 2011 Tohoku Earthquake (Mw 9.0). In particular, we focus on analysis of the CID characteristics caused by acoustic waves in near-field by utilizing both GNSS data processing methods and raytracing technique.

Data Processing

GNSS signal data ($L_1 = 1575.42$ MHz and $L_2 = 1227.60$ MHz) on selected nominal days and March 11th, 2011 were obtained from total 40 GPS Earth Observation Network (GEONET) stations. Obtained GNSS observation data were input into Long-Term Ionospheric Anomaly Monitor (LTIAM) to estimate Total Electron Content (TEC) values which corresponds to L1 signal. Longer period variations in TEC time series caused by multiple-hour changes (due to elevation angle of line-of-sight and diurnal & seasonal variations) are removed using the 4th order butterworth filter with a passband ranging from 4mHz (above the acoustic cutoff frequency, 3.3mHz) to 16mHz (below the Nyquist frequency 16mHz). The frequency of the CIDs generated by the acoustic waves in near-field, produced at the source, is localized by the wavelet analysis. Also, the time-distance diagram, which shows the TEC perturbation amplitude as a function of time and horizontal distance from the source, is employed to compute the velocities of the CIDs. The computed velocities are validated by the raytracing technique.

Results and Analysis

This section summarizes the results from the analysis of the CIDs during the Tohoku event. Figure 1 shows time series of the filtered Vertical TEC (VTEC) data. The Tohoku earthquake was occurred at about 5.8 hour (UT) as denoted with vertical dashed line. The ionospheric perturbations were observed 10~15 minutes after the Earthquakes. Maximum amplitude of the CIDs caused by the Tohoku event as estimated as 2.3 TECU.

Figure 2 shows the wavelet analysis result of station P117 viewing PRN 15. The white dashed line means the occurrence time of the earthquake. A frequency band of 5.5 mHz – 6.5 mHz (red area) was localized 10 –

15 minutes after the event.

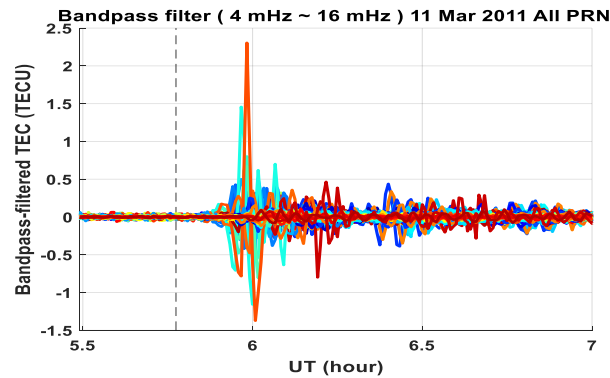


Figure 1 STEC observations for PRN 18 following the Tohoku earthquake (left). Filtered VTEC observations for all PRNs (right).

As shown in Figure 3 of a time-distance diagram, the black dashed line means expected slope of CIDs traveling 4 km/s (Rayleigh waves). The two black lines mean expected slope of CIDs traveling 1.18 km/s, 200 km horizontally apart from the epicenter, and 0.90 km/s, 400 km horizontally apart from the epicenter (acoustic waves in near-field). To validate these measured velocities of the acoustic wave-generated CIDs, we used the raytracing technique, the propagation profile of the acoustic waves from the source, (left of Figure 4). In this analysis, the thin shell of the ionosphere is assumed to be located at 350 km altitude. Thus, we can find the horizontal velocity profile of the CIDs at 350 km (right of the Figure 4) from the ray trace result. When the horizontal distances are 200 km and 400 km, the corresponding velocities are 1.07 km/s and 0.75 km/s. These results have a good agreement with the velocities of the CIDs computed from the time-distance diagram.

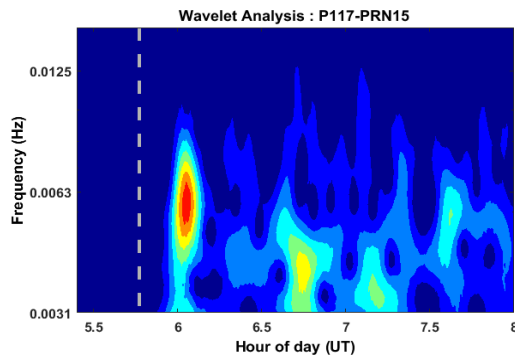


Figure 2 The wavelet analysis result of station P117 and PRN 15

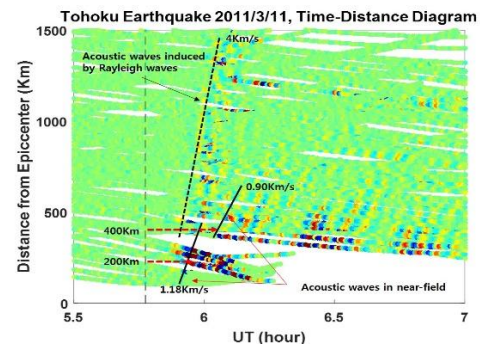


Figure 3 Time-distance Diagram

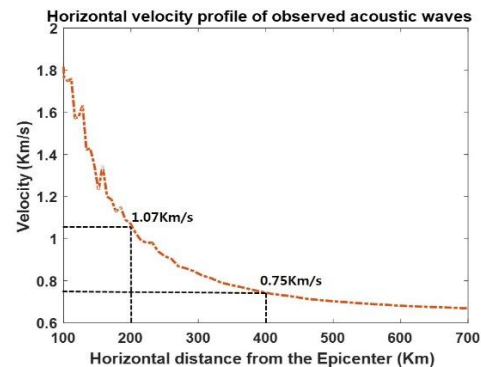
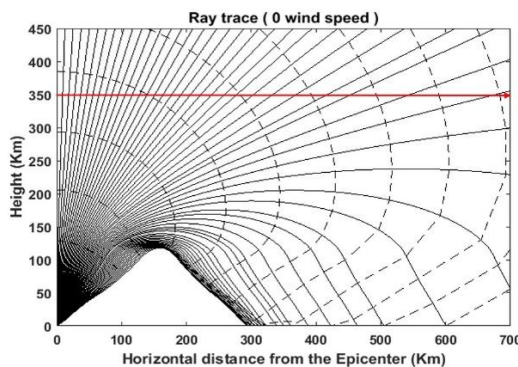


Figure 4 Ray traces of acoustic fields from a point source on the surface with no atmospheric wind (left) and Horizontal velocity profile from the ray trace result (right).